

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ-vs-frozen CI excludes 0
9 metrics × 8 encoders · WINNER col + BLUE BOX = surgery-vs-pretrain champion duel (tie = paired 95% CI overlaps 0) · 10 K bootstrap

	g0-frozen	g0-pre-2X	g0-pre-enc	g0-pre-hd	g0-s3DI-enc	g0-s3DI-hd	g0-sNoDI-enc	g0-sNoDI-hd	WINNER
action_top1 ↑ better	38.356 ±2.264	49.973 ±2.301*	50.027 ±2.329*	38.959 ±2.219	49.260 ±2.274*	50.247 ±2.301*	49.644 ±2.329*	48.712 ±2.301*	tie
motion_cos ↑ better	0.008 ±0.001	0.228 ±0.006*	0.113 ±0.004*	0.009 ±0.001*	0.121 ±0.004*	0.119 ±0.004*	0.124 ±0.004*	0.126 ±0.004*	ViT-g·2.0 pretrain_2X encoder
taxonomy_f1 ↑ better	0.778 ±0.017	0.759 ±0.018*	0.770 ±0.017	0.779 ±0.017	0.771 ±0.017	0.770 ±0.017	0.765 ±0.017	0.771 ±0.017	tie
future_mse ↓ better	1.645 ±0.002	1.784 ±0.003*	1.756 ±0.002*	1.646 ±0.002	1.718 ±0.002*	1.755 ±0.002*	1.715 ±0.002*	1.756 ±0.002*	ViT-g·2.0 pretrain head
rollout ↓ better	-0.008 ±0.000	-0.007 ±0.000*	-0.011 ±0.000*	-0.008 ±0.000	-0.006 ±0.000*	-0.011 ±0.000*	-0.007 ±0.000*	-0.011 ±0.000*	tie
causal ↓ better	1.656 ±0.002	1.776 ±0.003*	1.744 ±0.002*	1.656 ±0.002	1.721 ±0.002*	1.744 ±0.002*	1.719 ±0.002*	1.744 ±0.002*	ViT-g·2.0 pretrain head
tdist ↓ better	0.004 ±0.001	0.005 ±0.001*	0.001 ±0.001*	0.004 ±0.001	0.003 ±0.001*	0.001 ±0.001*	0.004 ±0.001	0.001 ±0.001*	tie
teacher_free ↓ better	0.023 ±0.000	0.032 ±0.000*	0.025 ±0.000*	0.023 ±0.000	0.042 ±0.001*	0.025 ±0.000*	0.039 ±0.001*	0.025 ±0.000*	ViT-g·2.0 pretrain head
maskratio ↓ better	0.052 ±0.001	0.070 ±0.001*	0.062 ±0.001*	0.052 ±0.001	0.060 ±0.001*	0.062 ±0.001*	0.061 ±0.001*	0.062 ±0.001*	ViT-g·2.0 pretrain head